

PYKV: A Scalable In-Memory Key-Value Store with Persistence

Milestone 4

◆ ABSTRACT

This milestone focuses on completing the PyKV system by adding replication, client interaction, and performance testing. The system is improved to support communication between servers and allow users to access it remotely using a client.

◆ STATEMENT PROBLEM

A single server system may fail and lose data. To solve this, replication and client-server communication are implemented to improve reliability and usability.

◆ MODULES USED

Replication Module

Client Module

Benchmarking Module

◆ WEEKLY PLAN

Week 7: Implement replication between servers

Week 8: Develop client and perform testing

◆ WORK DONE

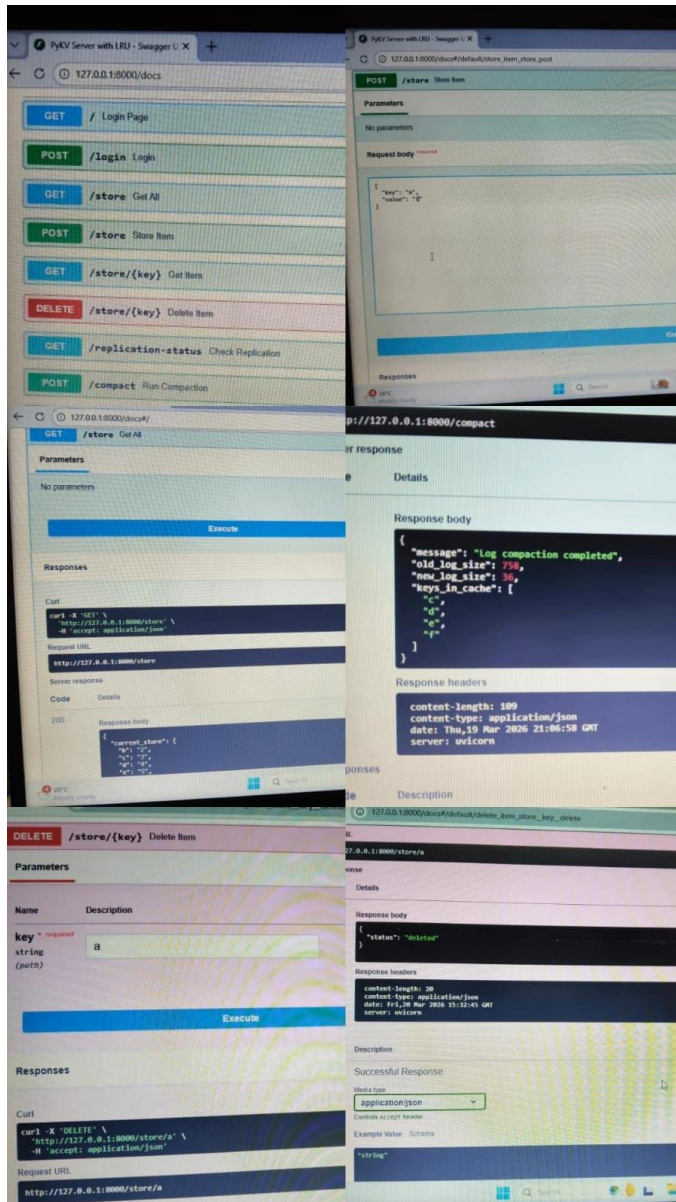
Implemented replication between primary and replica servers to maintain data backup. Developed a command-line client to send GET, SET, and DELETE requests to the server. Performed basic performance testing to check system speed and response time.

◆ PROBLEMS FACED

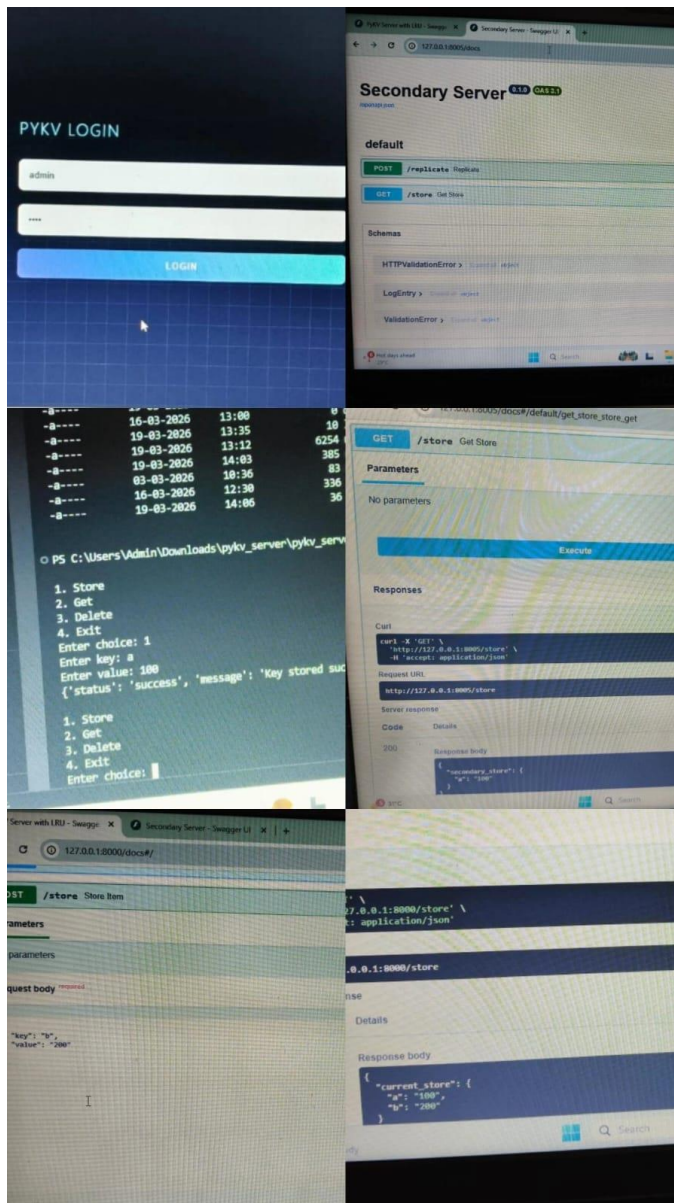
Faced issues in replication and client-server communication. These were solved by correcting API requests and testing properly.

◆ OUTPUT SCREENSHOT

Primary Server



Replication server



◆ CONCLUSION

Successfully completed the PyKV project with replication and client-server communication. The system works efficiently and handles operations correctly